Client Brief: Early Identification of At-Risk Writers in Primary Education

Client: Data2Intel Learning Analytics
Sector: Primary Education (K–6), Australia
Engagement type: Predictive analytics consulting
Data scope: 2,000 students across 40+ primary schools (2016–2020)

Background

Data2Intel is an Australian learning analytics consultancy serving a consortium of over forty primary schools. The consortium wants to identify students at risk of underperforming in writing before they reach the Year 3 NAPLAN assessment, the first national benchmark for literacy and numeracy.

Research shows that foundational skills acquired in the early years (Kindergarten to Year 2, ages 5–7) are strong predictors of long-term academic and professional outcomes. Currently, intervention happens reactively, after poor NAPLAN results are already in. The consortium wants to shift to a proactive model: flag at-risk students early enough for targeted support to make a difference.

Objective

Build a predictive model that identifies students likely to underperform in Year 3 writing, using reading, numeracy, and demographic data collected during Year 1 and Year 2. The model must be interpretable enough for educators to act on and defensible enough for Data2Intel to deploy across its school network.

Dataset

A curated dataset of 2,000 student records spanning five cohort years, including:

- **Early literacy indicators:** Burt Reading scores, Writing Vocabulary assessments, and letter identification results collected at the start and end of Year 1 and Year 2
- **Numeracy indicators:** formative numeracy assessment scores across the same time points
- **Demographics:** socioeconomic status (SES), school identifiers, gender
- **Disability flags:** recorded disability conditions
- **Target variable:** `Year3_Writing_At_Risk`, a binary label derived from Year 3 NAPLAN writing results

Assessments in this age group are not traditional pen-and-paper tests; they include dialogue and interview-based evaluations administered by trained teachers. No missing data; incomplete records were removed during curation.

Deliverables

1. **Exploratory analysis** addressing six specific questions from the client:
    - SES distribution across the student population in Year 1 and Year 2, benchmarked against national averages
    - Reading skill trajectories (start and end of Year 1 and Year 2)
    - Relationship between early writing vocabulary and Year 3 writing risk
    - Correlation between literacy skills, numeracy skills, and Year 3 writing risk
    - Disability conditions and their association with writing risk
    - Any additional insights that could inform early intervention design
2. **Two supervised predictive models** for `Year3_Writing_At_Risk`, compared and evaluated with a recommended production model
3. **One clustering model** to identify natural groupings within the student population that may reveal intervention-relevant segments
4. **Ethical and legal considerations** for deploying predictive models in a primary education context involving minors

Stakeholders

Role	Name	Needs
Director of Data & Insights	Dr. Alok Sinha	Technical depth: model architecture, performance metrics, deployment considerations
Director of Education & Engagement	Sally Tran	Actionable insights: what do the findings mean for schools and teachers?

Constraints & Considerations

- The model will be used to allocate limited intervention resources, so **recall for at-risk students matters more than overall accuracy** because missing a struggling student is costlier than a false alarm
- Predictions involve children aged 5–7; ethical handling of disability data, SES, and demographic features is non-negotiable
- The consortium includes both Catholic system and independent schools with different SES profiles (national averages of 100 and 102 respectively in 2018)
- Any deployed model must be explainable to non-technical educators who will act on its outputs